Hyperbolic Functions

→ The quantity $\frac{e^{y}-e^{-y}}{2}$, where y being real or complex is called hyperbolic sin y written as sinh y.

i.e, $\sinh y = \frac{e^{y}-e^{-y}}{2}$

→ Illy the quantity $\frac{e^{y}+e^{-y}}{2}$ is called hyperbolic cosine of y and is written as cosh y.

i.e, $\cosh y = \frac{e^{y}+e^{-y}}{2}$

* $\tanh y = \frac{\sinh y}{\cosh y} = \frac{e^{y}-e^{-y}}{e^{y}+e^{-y}}$ * $\operatorname{cosech} y = \frac{2}{e^{y}-e^{-y}}$

* $\coth y = \frac{e^{y}+e^{-y}}{e^{y}-e^{-y}}$ * $\operatorname{sech} y = \frac{2}{e^{y}+e^{-y}}$

Note

$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$

$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$

Q. P.T $\cos iy = \cosh y$?

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$\cos(iy) = \frac{e^{i(iy)} + e^{-i(iy)}}{2}$$

$$= \frac{e^{-y} + e^{y}}{2} = \cosh y$$

✓ Q. P.T ~~sinhy~~ $\sin(iy) = i\sinh y$?

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\sin(iy) = \frac{e^{i(iy)} - e^{-i(iy)}}{2i}$$

$$= \frac{e^{-y} - e^{y}}{2i}$$

$$= \frac{-i(e^{-y} - e^{y})}{2}$$

$$= \frac{i(e^{y} - e^{-y})}{2}$$

$$= i\sinh y$$

✓ Q. P.T $\tan iy = i\tanh y$?

$LHS = \tan iy = \frac{\sin iy}{\cos iy} = \frac{i \sinh y}{\cosh y} = i \tanh y$

Q. P.T $\cosh^2 x - \sinh^2 x = 1$?

$LHS = \cosh^2 x - \sinh^2 x$

$= \left(\frac{e^{y}+e^{-y}}{2}\right)^2 - \left(\frac{e^{y}-e^{-y}}{2}\right)^2$

$= \frac{e^{2y} + 2e^{y}.e^{-y} + e^{-2y} - e^{2y} + 2.e^{y}.e^{-y} - e^{-2y}}{4}$

$= 4/4 = 1$

OR

$\cos^2\theta + \sin^2\theta = 1$

replace θ by ix

$\cos^2 ix + \sin^2 ix = 1$

$(\cos ix)^2 + (\sin ix)^2 = 1$

$(\cosh x)^2 + (i \sinh x)^2 = 1$

$\cosh^2 x - \sinh^2 x = 1$

Q. P.T $\cosh 2x = 1 + 2\sinh^2 x$

$\cos 2\theta = 1 - 2\sin^2\theta$

replace θ by ix

$\cos(2ix) = 1 - 2\sin^2(ix)$

$\cos 2hx = 1 - 2(i\sinh x)^2$

$= 1 + 2\sinh^2 x$

Q. P.T $\cosh 2x = 2\cosh^2 x - 1$

$\cos 2\theta = 2\cos^2\theta - 1$

$\cos(2ix) = 2(\cos ix)^2 - 1$

$\cos h2x = 2\cosh^2 x - 1$

Q. P.T $\sinh 3x = 3\sinh x + 4\sinh^3 x$

$\cosh 3x = 4\cosh^3 x - 3\cosh x$.

Q. P.T $\tanh 3x = \dfrac{3\tanh x + \tanh^3 x}{1 + 3\tanh^2 x}$

$\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$

replace θ by ix

$\tan 3ix = \dfrac{3\tan ix - \tan^3 ix}{1 - 3\tan^2 ix}$

$\tanh 3x = \dfrac{3\tanh x - i^3\tanh^3 x}{1 - 3\times i^2\tanh^2 x} = \dfrac{3\tanh x + i\tanh^3 x}{1 + 3\tanh^2 x}$

Q. P.T: $\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$

$\sin(A+B) = \sin A \cos B + \cos A \sin B$

replace A by ix and B with iy.

$\sin(ix+iy) = \sin ix . \cos iy + \cos ix \sin iy$

$\sin i(x+y) = i\sinh x . \cosh y + \cosh x . i\sinh y$

$i\sinh(x+y) = i(\sinh x . \cosh y + \cosh x . \sinh y)$

$\Rightarrow \sinh(x+y) = \sinh x . \cosh y + \cosh x . \sinh y$

Q. P.T $\tan(\alpha+\beta) = \dfrac{\tanh\alpha + \tanh\beta}{1+\tanh\alpha . \tanh\beta}$

$\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$

replace A by $i\alpha$ and B by $i\beta$

$\tan(i\alpha + i\beta) = \dfrac{\tan i\alpha + \tan i\beta}{1 - \tan i\alpha . \tan i\beta}$

$\tan i(\alpha+\beta) = \dfrac{i\tanh\alpha + i\tanh\beta}{1 - i\tanh\alpha . i\tanh\beta}$

$i\tanh(\alpha+\beta) = \dfrac{i(\tanh\alpha + \tanh\beta)}{1+\tanh\alpha . \tanh\beta}$

Imp Q. If $v = \log\left[\tan\left(\pi/4 + \theta/2\right)\right]$, P.T $\tanh\frac{v}{2} = \tan\left(\frac{\theta}{2}\right)$?

Given $v = \log\left(\tan\left(\pi/4 + \theta/2\right)\right)$

Taking exponential on both sides,

$$e^{v} = \tan\left(\pi/4 + \theta/2\right)$$

$$= \frac{\tan\pi/4 + \tan\theta/2}{1 - \tan\pi/4 \cdot \tan\theta/2}$$

$$= \frac{1 + \tan\theta/2}{1 - \tan\theta/2}$$

Now, LHS $= \tanh v/2$

$$= \frac{e^{v/2} - e^{-v/2}}{e^{v/2} + e^{-v/2}} \quad (\text{x and} \div \text{by } e^{v/2})$$

$$= \left(\frac{e^{v/2} - e^{-v/2}}{e^{v/2} + e^{-v/2}}\right) \times \frac{e^{v/2}}{e^{v/2}}$$

$$= \frac{e^{v} - 1}{e^{v} + 1} = \frac{\frac{1 + \tan\theta/2}{1 - \tan\theta/2} - 1}{\frac{1 + \tan\theta/2}{1 - \tan\theta/2} + 1}$$

$$= \frac{1 + \tan\theta/2 - 1 + \tan\theta/2}{1 + \tan\theta/2 + 1 - \tan\theta/2} = \frac{2\tan\theta/2}{2} = \underline{\tan\theta/2}$$

PERIODS OF Hyperbolic Functions

$\cosh(x+iy) = \cos i(x+iy)$

$= \cos(ix + i^2 y)$

$= \cos(ix - y)$

$= \cos(ix - y - 2\pi)$

$= \cos(ix + i^2 y + i^2 2\pi)$

$= \cos i(x + iy + i2\pi)$

Hence hyperbolic ~~functions~~ cosine are periodic with period being imaginary i.e, $i2\pi$.

Illy we can show that period of $\sinh(x+iy)$ is $i2\pi$ and $\tanh(x+iy)$ is πi.

Separation into Real and Imaginary Parts.

Q. Separate the following into real and imaginary parts?

i) $\sin(\alpha + i\beta)$

$\sin(\alpha+i\beta) = \sin\alpha \cos i\beta + \cos\alpha \sin i\beta$

$= \sin\alpha \cosh\beta + i(\cos\alpha \sinh\beta)$

ii) $\cos(\alpha + i\beta)$

$\cos(\alpha + i\beta) = \cos\alpha \cos i\beta - \sin\alpha \sin i\beta$

$= \cos\alpha \cdot \cosh\beta - \sin\alpha \times i\sinh\beta$

$= \cos\alpha \cdot \cosh\beta - i(\sin\alpha \sinh\beta)$

iii) $\tan(\alpha+i\beta) = \frac{2}{2} \frac{\sin(\alpha+i\beta)}{\cos(\alpha+i\beta)} \times \frac{\cos(\alpha-i\beta)}{\cos(\alpha-i\beta)}$

$\sin\alpha + \sin\beta$
$= 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$

$$= \frac{\sin(\alpha+i\beta+\alpha-i\beta) + \sin(\alpha+i\beta-\alpha+i\beta)}{\cos(\alpha+i\beta+\alpha-i\beta) + \cos(\alpha+i\beta-\alpha+i\beta)}$$

$$= \frac{\sin(2\alpha) + \sin 2i\beta}{\cos 2\alpha + \cos(2i\beta)}$$

$\sin\alpha\cos\beta = \frac{1}{2}\left[\sin(\alpha+\beta) + \sin(\alpha-\beta)\right]$

$\cos\alpha\cos\beta = \frac{1}{2}\left[\cos(\alpha+\beta) + \cos(\alpha-\beta)\right]$

$$= \frac{\sin 2\alpha + i\sinh 2\beta}{\cos 2\alpha + \cosh 2\beta}$$

$$= \frac{\sin 2\alpha}{\cos 2\alpha + \cosh 2\beta} + i\,\frac{\sinh 2\beta}{\cos 2\alpha + \cosh 2\beta}$$

Hw i) $\cot(\alpha+i\beta)$ ii) $\sec(\alpha+i\beta)$ iii) $\operatorname{cosec}(\alpha+i\beta)$

iv) $\cosh(\alpha+i\beta) = \cos i(\alpha+i\beta)$

$= \cos(\alpha i - \beta)$

$= \cos i\alpha \cos\beta + \sin i\alpha \sin\beta$

$= \cosh\alpha\cos\beta + i\sinh\alpha\sin\beta$

v) $\sinh(\alpha+i\beta) = -i\sin i(\alpha+i\beta)$

$= -i(\sin(\alpha i-\beta))$

$= -i\left[\sin i\alpha\cos\beta - \cos i\alpha\sin\beta\right]$

$= \cancel{-i\sin i\alpha}$

$= -i\left[i\sinh\alpha\cos\beta - \cosh\alpha\sin\beta\right]$

$= \sinh\alpha\cos\beta + i\cosh\alpha\sin\beta$

Q. $\tanh(\alpha+i\beta) = -\tan i(\alpha+i\beta)$

$= -\tan(\alpha i-\beta)$

$= -\dfrac{\sin(\alpha i-\beta)}{\cos(\alpha i-\beta)}$

$= -i\left[\dfrac{2\sin(i\alpha-\beta)\cos(i\alpha+\beta)}{2\cos(i\alpha-\beta)\cos(i\alpha+\beta)}\right]$

$= -i \times \dfrac{\sin(i\alpha-\beta+i\alpha+\beta)+\sin(i\alpha-\beta-i\alpha-\beta)}{\cos(i\alpha-\beta+i\alpha+\beta)+\cos(i\alpha-\beta-i\alpha-\beta)}$

$= -i\left[\dfrac{\sin 2i\alpha - \sin 2\beta}{\cos 2i\alpha + \cos 2\beta}\right]$

$= -i\left[\dfrac{i\sinh 2\alpha - \sin 2\beta}{\cosh 2\alpha + \cos 2\beta}\right] = \dfrac{\sinh 2\alpha - i\sin 2\beta}{\cosh 2\alpha + \cos 2\beta}$

$= \sinh 2\alpha / \cosh 2\alpha + \cos 2\beta - i\sin 2\beta / \cosh 2\alpha + \cos 2\beta$

Q. P.T $\tan\left(\frac{u+iv}{2}\right) = \frac{\sin u + i\sinh v}{\cos v + \cosh v}$?

$$LHS = \tan\left(\frac{u+iv}{2}\right)$$

$$= \frac{\sin\left(\frac{u+iv}{2}\right)}{\cos\left(\frac{u+iv}{2}\right)}$$

$$= \frac{2\sin\left(\frac{u+iv}{2}\right).\cos\left(\frac{u-iv}{2}\right)}{2\cos\left(\frac{u+iv}{2}\right).\cos\left(\frac{u-iv}{2}\right)}$$

$$= \frac{\sin\left(\frac{u+iv}{2}+\frac{u-iv}{2}\right) + \sin\left(\frac{u+iv}{2} - \left(\frac{u-iv}{2}\right)\right)}{\cos\left(\frac{u+iv}{2}+\frac{u-iv}{2}\right) + \cos\left(\frac{u+iv}{2} - \left(\frac{u-iv}{2}\right)\right)}$$

$$= \frac{\sin v + \sin iv}{\cos v + \cos iv}$$

$$= \frac{\sin u + i\sinh v}{\cos v + \cosh v}$$

Q. If $\sin(A+iB) = x+iy$, P.T

i) $\frac{x^2}{\cosh^2 B} + \frac{y^2}{\sinh^2 B} = 1$ or $x^2\operatorname{sech}^2 B + y^2\operatorname{cosech}^2 B = 1$

ii) $\frac{x^2}{\sin^2 A} - \frac{y^2}{\cos^2 A} = 1$ or $x^2 \operatorname{cosec}^2 A - y^2 \sec^2 A = 1$

Given $\sin(A+iB) = x+iy$

$\sin A \cos iB + \cos A \sin iB = x+iy$

$\sin A \cosh B + i \cos A \sinh B = x+iy$

$\Rightarrow x = \sin A \cosh B, \quad y = \cos A \sinh B.$

i) $\frac{x^2}{\cosh^2 B} + \frac{y^2}{\sinh^2 B} = \frac{\sin^2 A \cosh^2 B}{\cosh^2 B} + \frac{\cos^2 A \sinh^2 B}{\sinh^2 B}$

$= \sin^2 A + \cos^2 A = \underline{\underline{1}}$

ii) $\frac{x^2}{\sin^2 A} - \frac{y^2}{\cos^2 A} = \frac{\sin^2 A \cosh^2 B}{\sin^2 A} - \frac{\cos^2 A \sinh^2 B}{\cos^2 A}$

$= \cosh^2 B - \sinh^2 B$

$= \underline{\underline{1}}$

Q. If $\tan(A+iB) = x+iy$, P.T i) $x^2+y^2+2x\cot 2A = 1$

ii) $x^2+y^2-2y\coth 2B+1=0$

Given $\tan(A+iB) = x+iy \Rightarrow \tan(A-iB) = x-iy$

$\tan 2A = \tan(A+iB+A-iB) = \tan\left((A+iB)+(A-iB)\right)$

$= \frac{\tan(A+iB) + \tan(A-iB)}{1 - \tan(A+iB).\tan(A-iB)}$

$$= \frac{x+iy+x-iy}{1-(x+iy)(x-iy)}$$

$$= \frac{2x}{1-(x^2+y^2)} = \frac{2x}{1-x^2-y^2}$$

$$\Rightarrow 1-x^2-y^2 = \frac{2x}{\tan 2A} = 2x.\cot 2A$$

$$\Rightarrow x^2+y^2+2x\cot 2A = 1$$

(ii) $\tan 2iB = \tan\left(A+iB-(A-iB)\right)$

$$= \frac{\tan(A+iB)-\tan(A-iB)}{1+\tan(A+iB).\tan(A-iB)}$$

$$= \frac{(x+iy)-(x-iy)}{1+(x+iy)(x-iy)}$$

$$= \frac{2iy}{1+x^2+y^2}$$

$$\Rightarrow 1+x^2+y^2 = \frac{2iy}{\tan 2iB} = \frac{2iy}{i\tanh 2B} = 2y\coth 2B$$

$$\Rightarrow x^2+y^2-2y\coth 2B+1 = 0$$

Q. If $\sin(\theta+i\phi) = \cos\alpha + i\sin\alpha$, P.T. $\cos^2\theta = \pm\sin\alpha$?

$\sin(\theta+i\phi) = \sin\theta\cos i\phi + \cos\theta\sin i\phi = \cos\alpha + i\sin\alpha$

$= \sin\theta\cosh\phi + i\cos\theta\sinh\phi = \cos\alpha + i\sin\alpha$

$$\Rightarrow \sin\theta\cosh\phi = \cos\alpha, \quad \cos\theta\sinh\phi = -\sin\alpha$$

$$\cosh\phi = \frac{\cos\alpha}{\sin\theta}, \quad \sinh\phi = \frac{\sin\alpha}{\cos\theta}$$

$$\cosh^2\phi - \sinh^2\phi = 1$$

$$\frac{\cos^2\alpha}{\sin^2\theta} - \frac{\sin^2\alpha}{\cos^2\theta} = 1$$

$$\cos^2\alpha \cdot \cos^2\theta - \sin^2\alpha \cdot \sin^2\theta = \sin^2\theta \cdot \cos^2\theta$$

$$(1-\sin^2\alpha)\cdot\cos^2\theta - \sin^2\alpha\,(1-\cos^2\theta) = (1-\cos^2\theta)\cdot\cos^2\theta$$

$$\cancel{\cos^2\theta} - \cancel{\cos^2\theta\cdot\sin^2\alpha} - \sin^2\alpha + \cancel{\sin^2\alpha\cos^2\theta} = \cancel{\cos^2\theta} - \cos^4\theta$$

$$-\sin^2\alpha = -\cos^4\theta$$

$$\sin^2\alpha = \cos^4\theta$$

$$\therefore \cos^2\theta = \pm\sin\alpha$$

Q. If $A + iB = C\tan(x+iy)$, P.T. $\tan 2x = \dfrac{2AC}{C^2 - A^2 - B^2}$?

$$\tan(x+iy) = \frac{A+iB}{C} = A/C + i\,B/C$$

$$\tan(x-iy) = A/C - iB/C$$

$$\tan 2x = \tan\big((x+iy)+(x-iy)\big)$$

$$= \frac{\tan(x+iy) + \tan(x-iy)}{1 - \tan(x+iy)\cdot\tan(x-iy)}$$

$$= \frac{A/C + iB/C + A/C - iB/C}{1 - (A/C + iB/C)(A/C - iB/C)}$$

$$= \frac{2A/c}{1-\left(A^2/c^2+B^2/c^2\right)}$$

$$= \frac{2A/c}{\frac{c^2-A^2-B^2}{c^2}}$$

$$= \frac{2AC}{C^2-A^2-B^2}$$

Q. If $\tan(\theta+i\phi) = \sin(x+iy)$, P.T $\coth y \sinh 2\phi = \cot x \sin 2\theta$?

$$\tan(\theta+i\phi) = \sin(x+iy)$$

$$\frac{\sin(\theta+i\phi)}{\cos(\theta+i\phi)} = \frac{2\sin(\theta+i\phi)\cos(\theta-i\phi)}{2\cos(\theta+i\phi)\cdot\cos(\theta-i\phi)}$$

$$= \frac{\sin(\theta+i\phi+\theta-i\phi)+\sin(\theta+i\phi-\theta+i\phi)}{\cos(\theta+i\phi+\theta-i\phi)+\cos(\theta+i\phi-\theta-i\phi)}$$

$$= \frac{\sin 2\theta+\sin i2\phi}{\cos 2\theta+\cos i2\phi} = \frac{\sin 2\theta + i\sinh 2\phi}{\cos 2\theta+\cosh 2\phi} \quad \text{—} ①$$

$$\sin(x+iy) = \sin x \cos iy + \cos x \sin iy \quad \text{—} ②$$

$$= \sin x \cosh y + i\cos x \sinh y$$

$① = ② \Rightarrow$

$$\frac{\sin 2\theta}{\cos 2\theta+\cosh 2\phi} = \sin x \cosh y \quad ③, \qquad \frac{\sinh 2\phi}{\cos 2\theta+\cosh 2\phi} = \cos x \sinh y \quad ④$$

$$(3)/(4) \Rightarrow \frac{\sin 2\theta}{\sinh 2\phi} = \frac{\sin x \cosh y}{\cos x \sinh y}$$

$$= \tan x \coth y$$

$$\Rightarrow \coth y \sinh 2\phi = \sin 2\theta \cot x$$

HW

Q. If $\sin(\theta + i\phi) = \rho(\cos x + i \sin x)$, P.T

i) $\rho^2 = \frac{1}{2}(\cosh 2\phi - \cos 2\theta)$ ii) $\tan x = \tanh\phi - \cot\theta$

Inverse Hyperbolic Functions.

If $x = \sinh y$ and $y = \sinh^{-1} x$, then we say that y is the inverse hyperbolic sine of x. lly we can define $\cosh^{-1} x$, $\tanh^{-1} x$ etc.

Q. If x is real, S.T $\sinh^{-1} x = \log x + \sqrt{x^2+1}$?

Let $\sinh^{-1} x = y$

$$x = \sinh y = \frac{e^y - e^{-y}}{2}$$

$$2x = e^y - e^{-y} = e^y - 1/e^y$$

$$2xe^y = e^{2y} - 1$$

$$(e^y)^2 - 2xe^y - 1 = 0.$$

$$e^y = \frac{2x \pm \sqrt{4x^2+4}}{2} = x \pm \sqrt{x^2+1}$$